Prithul Das

Undergraduate Mechanical Engineering Student

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prithuldas
Prithul-Das

Education

Expected May 2027 — **B.Sc. in Mechanical Engineering**, *Ahsanullah University of Science and Technology (AUST)*, Dhaka, Bangladesh

Experience

2025 – Present — **Co-Founder**, *AUST Applied Modelling and Simulation Lab (AASML), AUST*, Dhaka, Bangladesh
- Co-founding a student-driven research lab alongside alumni, undergraduate students, and two faculty advisors.
- Initiating CFD-based research using ANSYS Fluent in aerodynamics and thermal-fluid systems while building a healthy research culture within the department.

Dec 2025 – Present — **Communication Director**, *IMechE AUST Student Chapter*, Dhaka, Bangladesh
- Leading official communications, technical event publicity, and professional correspondence.
- Coordinating outreach and documentation for engineering-focused events and seminars.

Jul 2025 – Dec 2025 — **Recruitment Director**, *IMechE AUST Student Chapter*, Dhaka, Bangladesh
- Led recruitment planning and candidate evaluation for new members.
- Assisted in team structuring and organizational onboarding.

Dec 2024 – Jul 2025 — **Junior Website and Publicity Director**, *IMechE AUST Student Chapter*, Dhaka, Bangladesh
- Managed event publicity and online presence for technical and professional programs.
- Supported hosting and coordination of engineering-related events.

2023 – Present — **Private Tutor (Mathematics & Physics)**, *Self-Employed*
- Tutored high school students in mathematics and physics.
- Developed structured problem sets and concept-focused study materials.

Research

- **Prithul Das**, et al., *Aerodynamic Comparison Analysis of Conic Rocket Fin Designs in Subsonic Flow*, Conference Paper, 2025. (Presented)
- **Prithul Das**, et al., *Experimental Performance Evaluation of an Earth Tunnel Heat Exchanger (ETHX) System for Space Cooling and Heating in Bangladesh*, Conference Paper, 2025. (Presented)

Technical Skills

Programming	Python, MATLAB, C++ (working knowledge)
CFD & Simulation	ANSYS Fluent, mesh generation, RANS turbulence modeling
Scientific Computing	NumPy, SciPy, Matplotlib
CAD	SolidWorks
Documentation	LaTeX (technical reports, conference papers, presentations)

Projects

Conic Rocket Fin Aerodynamic Analysis CFD-based comparison of straight, trapezoidal, sounder, and shark-caved fins in subsonic flow; resulted in a conference paper and presentation.

Earth Tunnel Heat Exchanger (ETHX) System Experimental investigation of a passive geothermal HVAC system for hot and humid climates; resulted in a conference paper and presentation.

Supersonic Flow Simulations Numerical simulations of compressible flow (Mach 6) to study shock structures and mesh sensitivity using ANSYS Fluent.

Relevant Coursework

Mathematics Multivariable Calculus, Differential Equations, Linear Algebra

Core Engineering Fluid Mechanics, Thermodynamics, Solid Mechanics

Computing Engineering Mathematics, Basic Programming